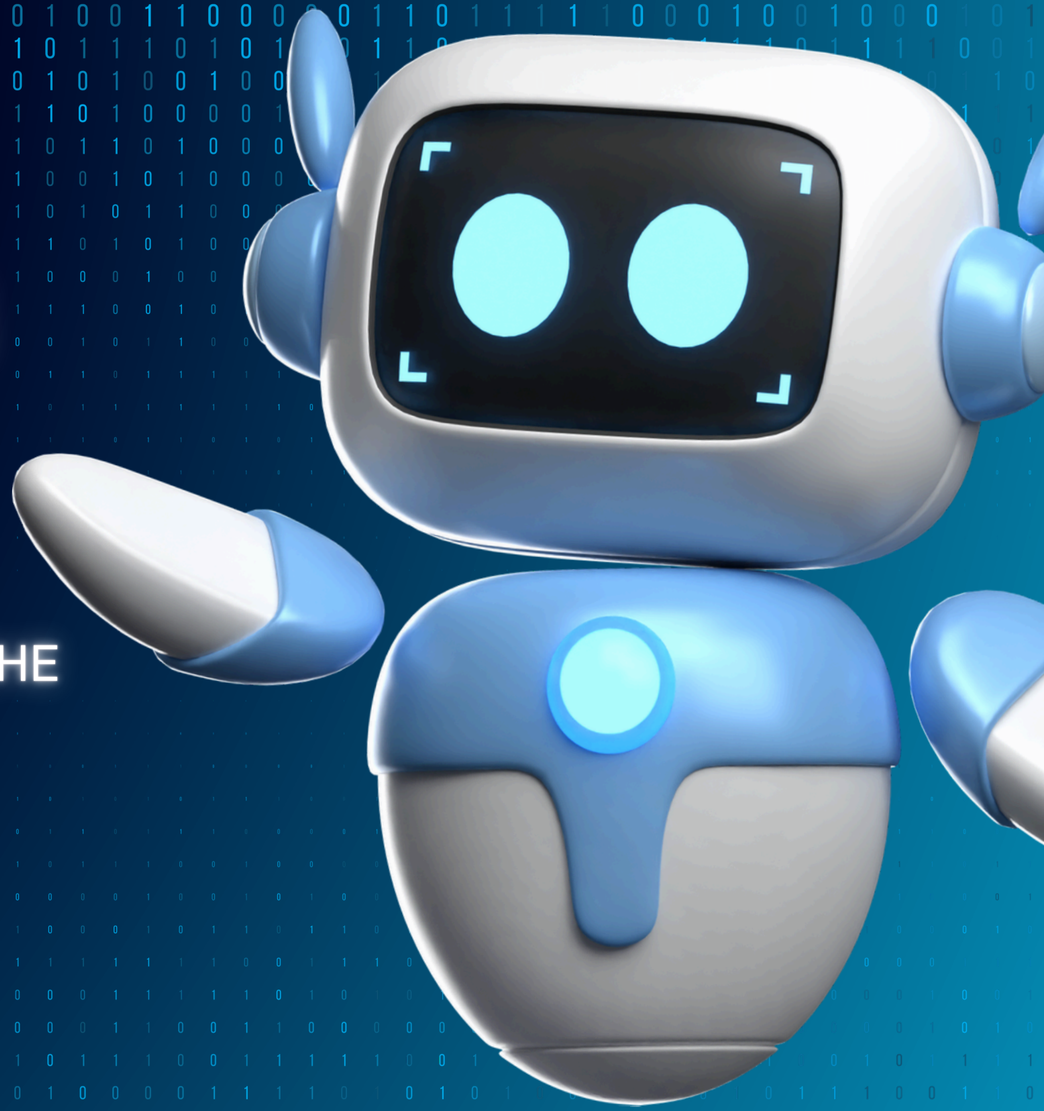
HATE SPEECH DETECTION USING TRANSFORMER MODELS



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1 WHAT IS A TRANSFORMER?

Transformer models are a type of neural network architecture designed for processing sequential data (text, images, audio) in parallel, rather than sequentially like RNNs. They excel at understanding context and relationships between data components using "attention" mechanisms.





HUGGING FACE

2 DATASET & PREPROCESSING

ARABIC

- 3 CSV files
- 458.277 rows
- 50% hate
- 50% non-hate

- Lowercasing
- Data Balancing
- Special character filtering
- Whitespace normalization
- URL, Mention , emoji & Hashtag removal



ENGLISH

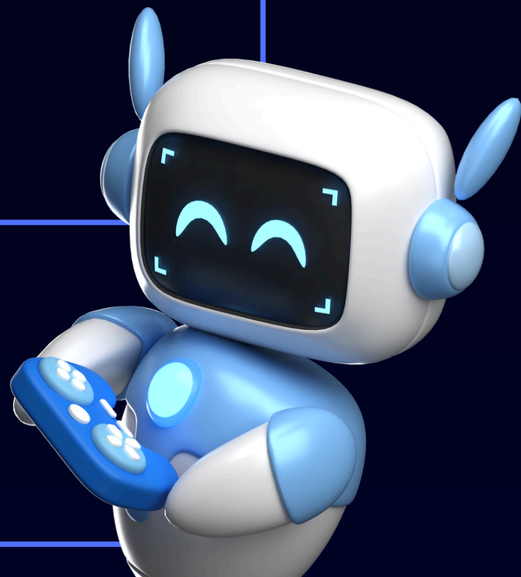
- 2 CSV files
- 711.174 rows
- 47% hate, 53% non-hate

FRENCH

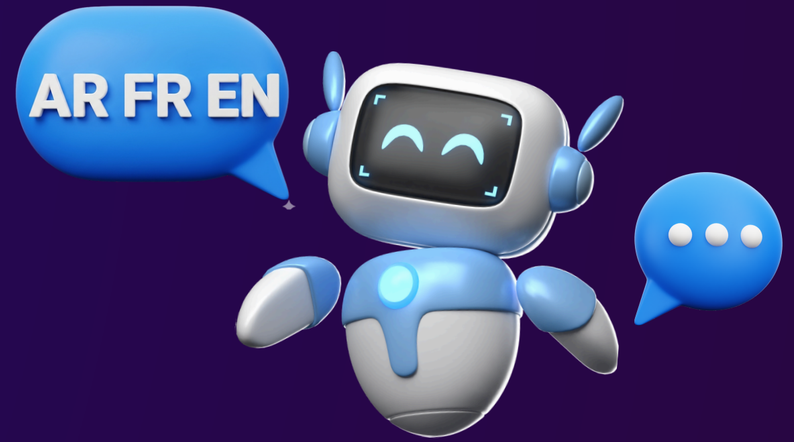
- 1 CSV File
- 28 164 Row
- 50% Hate
- 50% Not Hate

3 FINE-TUNING RESULTS

MODELS	BEFORE FINE-TUNING	AFTER FINE-TUNING
ROBERTA-BASE	• Accuracy : 0.8328 • F1 score : 0.8326 • Precision : 0.8328 • Recall : 0.8328	• Accuracy : 0.8702 • F1 score : 0.8701 • Precision : 0.8702 • Recall : 0.8702
QARIB	• Accuracy : 66.7% • Macro F1 : 0.5556	• Accuracy : 94% • Macro F1 : 0.9468
CAMEMBERT-V2	• Accuracy : 75% • Macro F1 : 0.6975	• Accuracy : 69% • Macro F1 : 0.6918



4 INFERENCE TESTS



12 sample/language

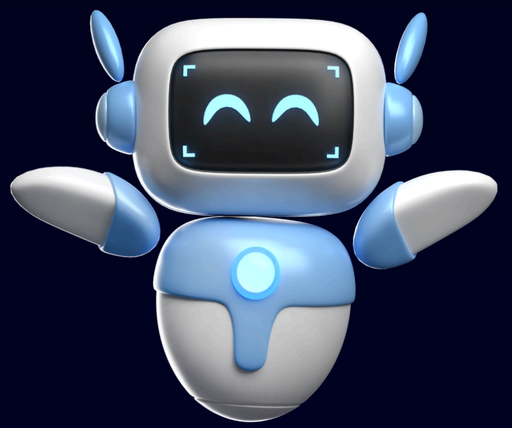
ENGLISH	33.33%	100%
ARABIC	66.66%	66.66%
FRENCH	75%	58.38%

5 POTENTIAL IMPROVEMENTS

Larger French dataset

Extended Hyperparameter search

Longer training for CamemBERTv2



THANK YOU FOR YOUR ATTENTION

